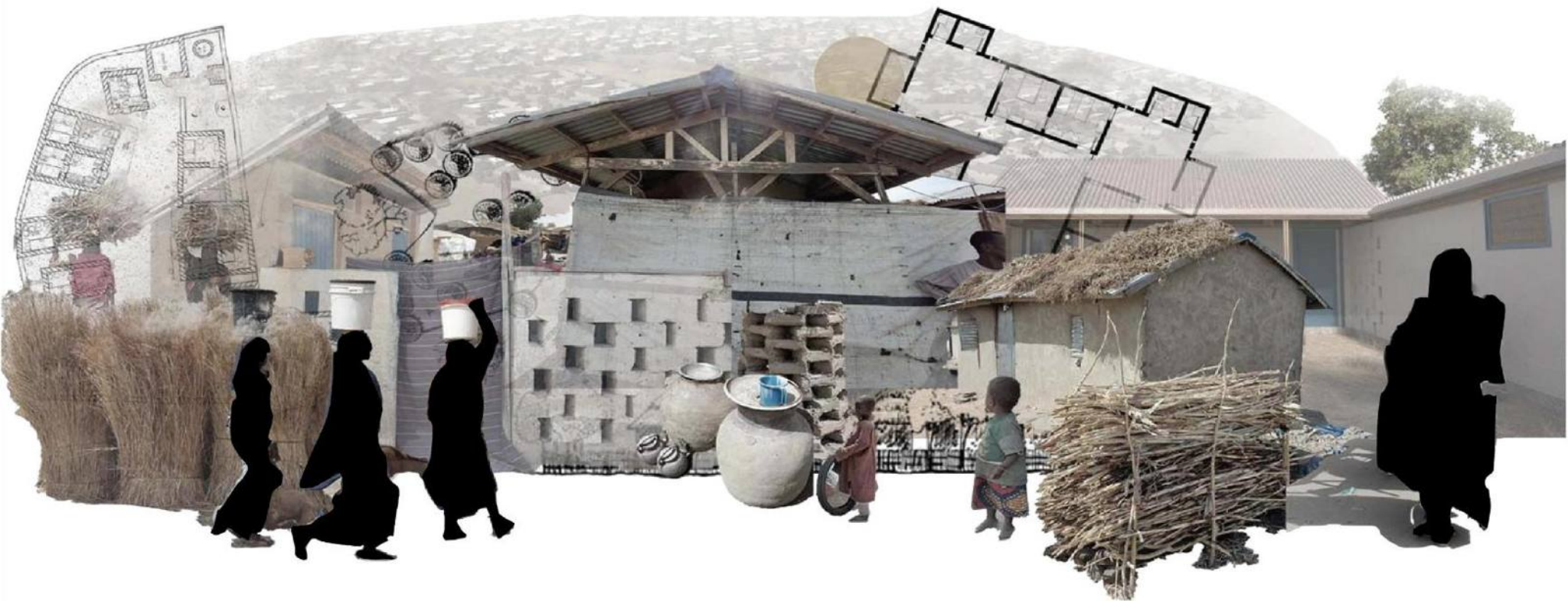


UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN



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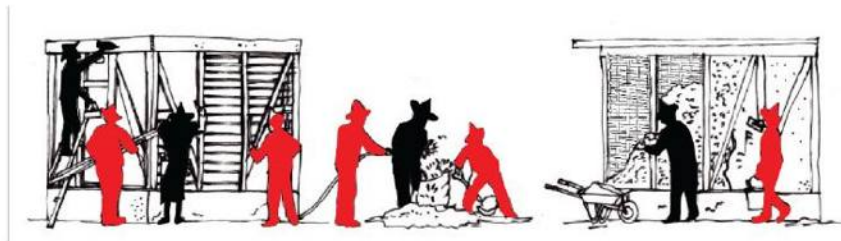


UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

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INTRODUCTION

- Background
- Problem Statement
- Objectives
- Study Rational
- Scope & Limitations

UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

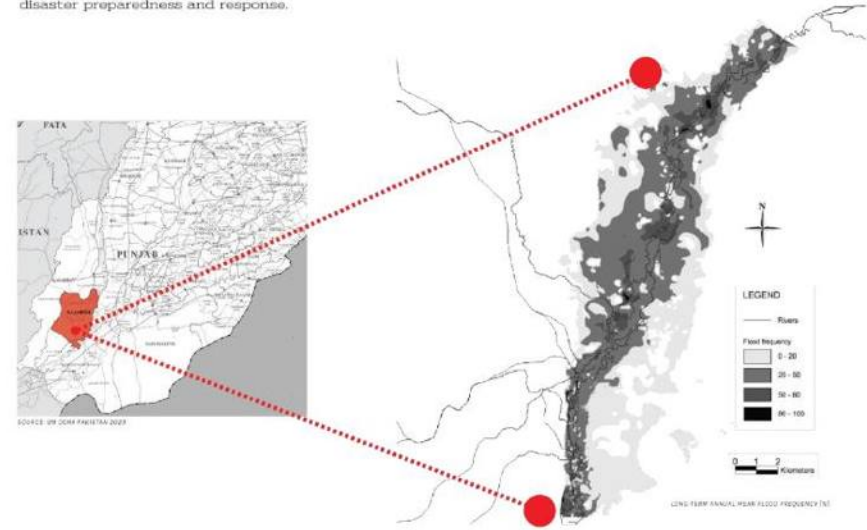
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SITE: HISTORY

Flood disasters have historically plagued Pakistan, with Punjab being one of the most vulnerable regions. This vulnerability is primarily due to its geography, which includes major rivers such as the Indus and its tributaries. These rivers, while essential for irrigation, frequently overflow during the monsoon season, leading to devastating floods. The district of Rajanpur, located near the Indus River and prone to hill torrents from the Suleiman Range, is particularly susceptible. The 2023 floods once again highlighted this risk, with Rajanpur experiencing severe damage to homes, agriculture, and infrastructure. The floods displaced 7535 people, with 2,170 individuals living in relief camps, and caused damage to 1980 houses, 8 schools, and 30 bridges across Pakistan between July and August 2024 (NDMA/PDMA).

Pakistan's susceptibility to floods stems from its unique topography and climatic conditions. The country experiences heavy monsoon rains, which often result in the overflowing of rivers and flash floods. These floods are exacerbated by inadequate infrastructure, deforestation, and poor urban planning, making disaster management a challenging task. Over the years, the government, along with humanitarian organizations, has implemented various flood mitigation measures, including embankments, early warning systems, and relief efforts. However, the persistent nature of floods highlights the need for continued investment in disaster preparedness and response.



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UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

(02)



SOURCE: ISLAMIC RELIEF DEP ANNUAL REPORT 2020

BACKGROUND

2010~2024

Between 2010 and 2024, Rajanpur, a district in southwestern Punjab, has been repeatedly affected by severe floods, primarily due to its proximity to the Indus River and the hill torrents from the Suleman Range. The devastating 2010 floods set a precedent for the challenges the district would continue to face, with subsequent floods in 2012, 2014, 2015, and 2022 causing widespread displacement, destruction of homes, and significant agricultural losses. Despite ongoing efforts to build embankments and improve flood defenses, Rajanpur remains highly vulnerable to these natural disasters.

In 2023, the pattern of devastation persisted as heavy monsoon rains once again led to severe flooding in the district. The floods displaced 7,535 people and caused extensive damage to 1,980 houses, 8 schools, and 30 bridges. These repeated disasters have highlighted the urgent need for more effective flood management strategies and infrastructure improvements to protect the residents and economy of Rajanpur from future floods.

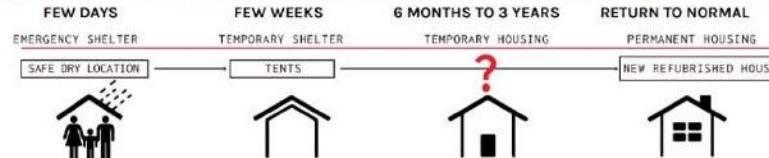
SITE ANALYSIS



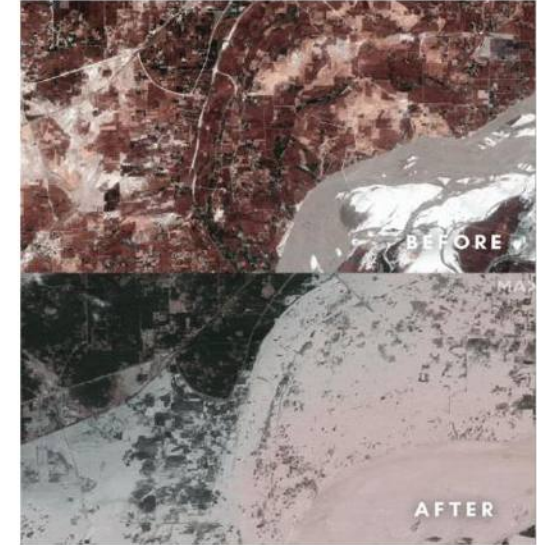
SOURCE: GOOGLE MAP

TIMELINE | From Disaster To Recovery

SOURCE: MULTIPLE SOURCES



PROBLEM | From Disaster To Recovery



SOURCE: ALJAZEERA 2023

STRATEGY | From Disaster To Recovery



SOURCE: NEWS WEBSITES

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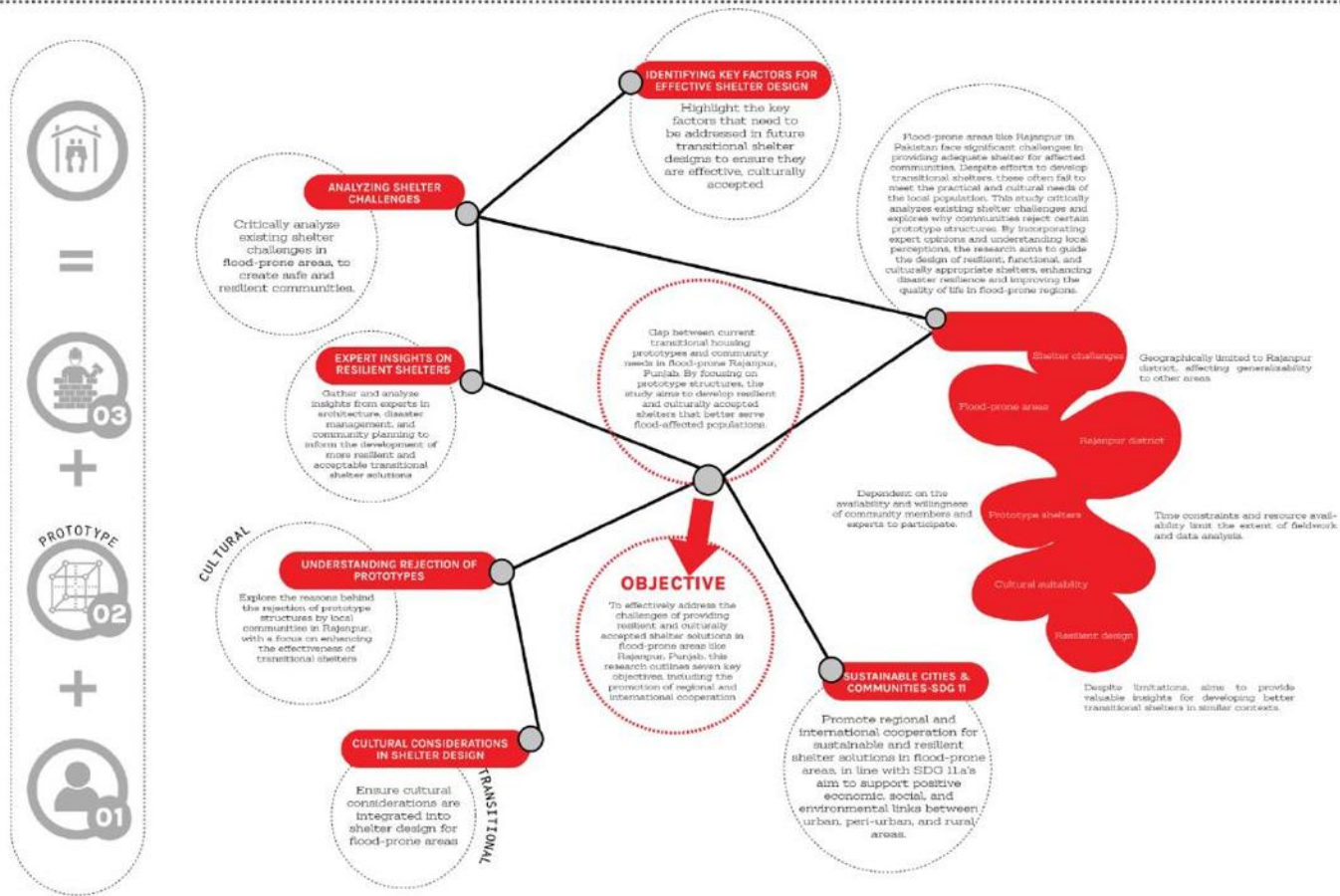
BACKGROUND

The 2022 Pakistan floods, which resulted in over 1,100 deaths and impacted more than 30 million people, underscore the nation's vulnerability to climate change (World Bank Group, 2022; GOP Finance Division, 2022). Rajanpur district, due to its proximity to the Indus River, was severely affected, with over 8 million people displaced and significant damage to housing and sanitation infrastructure (PDMA, 2022; REEDS, 2022). Despite the importance of transitional shelters in post-disaster recovery, the current District Disaster Management Plan for Rajanpur lacks adequate provisions for this critical phase, leading to prolonged displacement and poor living conditions (NDMA, 2016). The rejection of prototype shelters by local communities highlights the need for culturally sensitive and functional housing solutions. This research focuses on addressing these challenges by analyzing and improving prototype structures for transitional housing in Rajanpur, aiming to develop more effective and culturally accepted shelter solutions.

PROBLEM STATEMENT

There is a large void in the provision of effective shelter solutions in flood-prone areas of Punjab, Pakistan, particularly Rajanpur district. Communities are devastated by recurrent floods, which leave them vulnerable and exposed to weather. Nonetheless, it remains difficult to provide immediate and sustainable shelter despite the worthy attempts by both governmental and non-governmental organizations. The shortage of resources combined with geographical constraints only worsens this situation. This Research demonstrates that resilient and cost-effective solutions may be provided through the incorporation of local architectural practices into modern engineering approaches. This study aims to present innovative strategies for bridging the gap effectively.

The persistent rejection of transitional shelter prototypes by communities in flood-prone areas like Rajanpur highlights a significant disconnect between shelter design and community needs. Current shelters often fail to address key issues related to cultural suitability, functionality, and design. This study aims to identify and analyze these challenges, understand community concerns, and incorporate expert insights to propose effective solutions. The focus is on developing future shelter designs that are resilient, culturally accepted, and capable of providing safe and dignified living conditions for flood-affected populations.



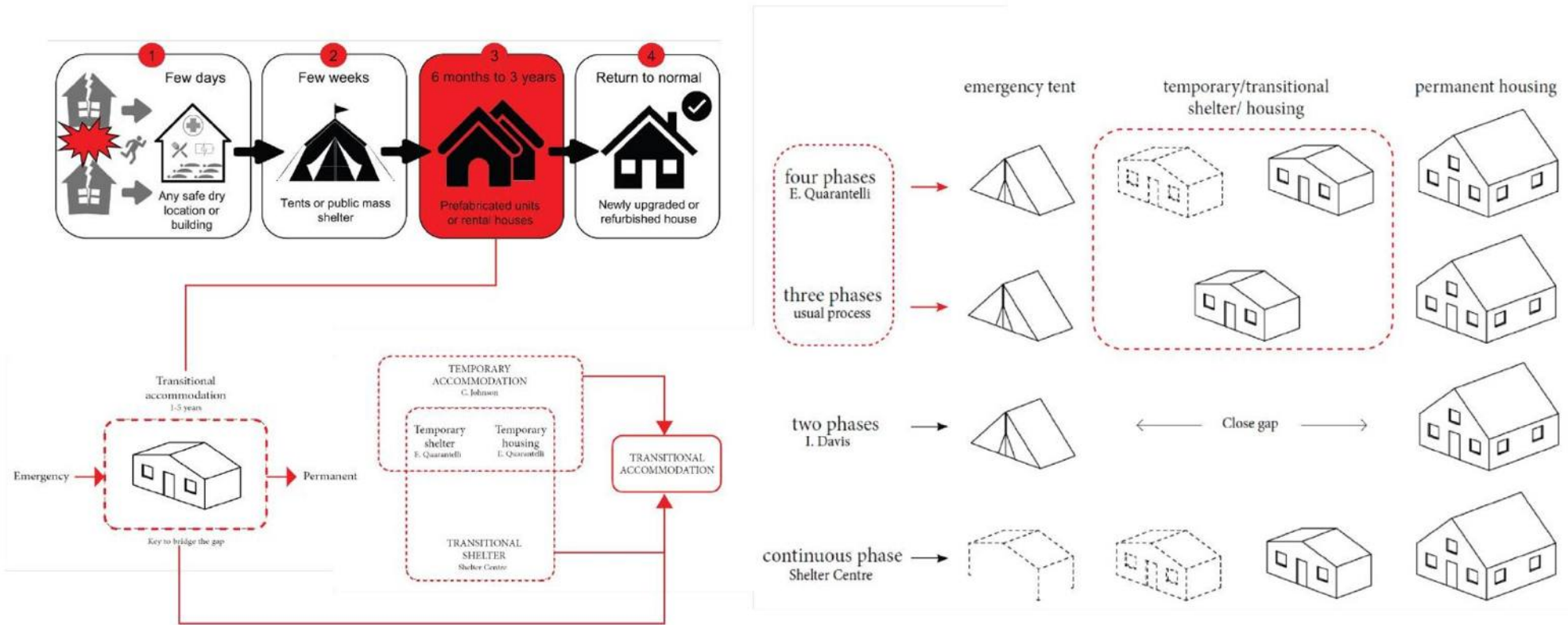
LITERATURE REVIEW

- Key Definition
- Existing Systematic Studies
- Existing Condition Anaylsis

UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

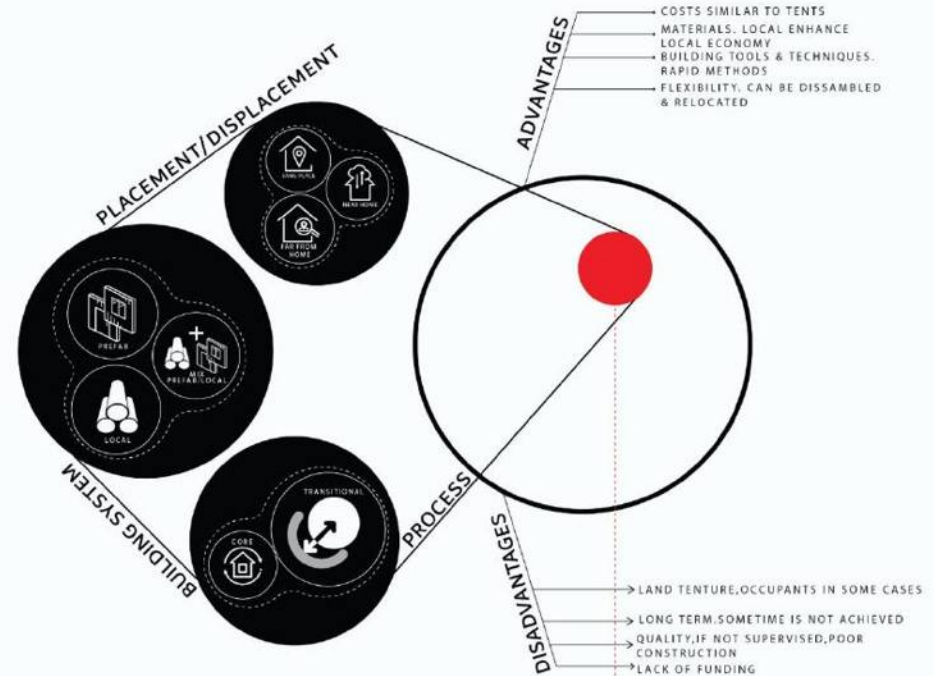
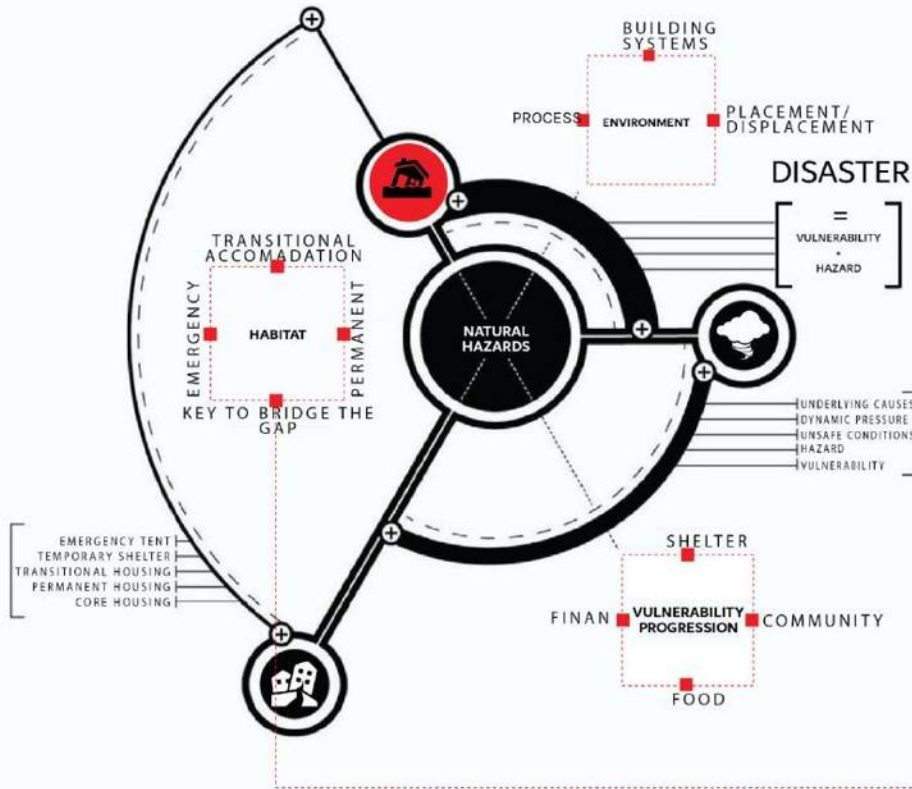
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UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

(06)



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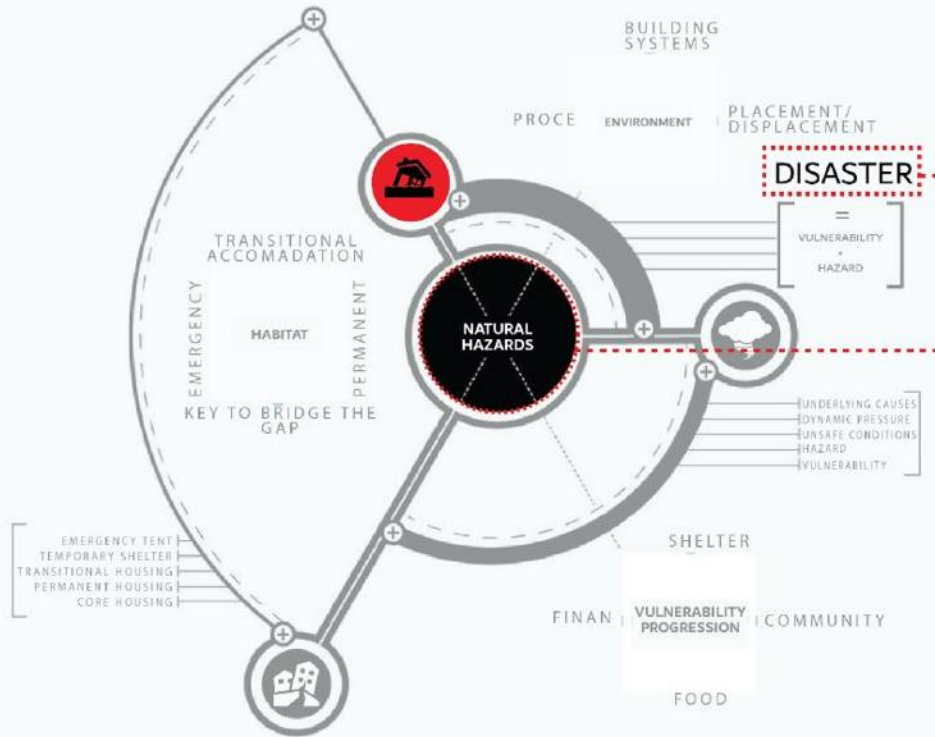
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UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

(06)



KEY WORD	REFERENCE	DEFINITION
DISASTER	DISASTER 2007	A significant disturbance to community or social operation due to hazardous events.

KEY WORD	REFERENCE	DEFINITION
NATURAL HAZARD	UNDRR 2022	A natural event such as a flood or earthquake that has the potential to cause damage.

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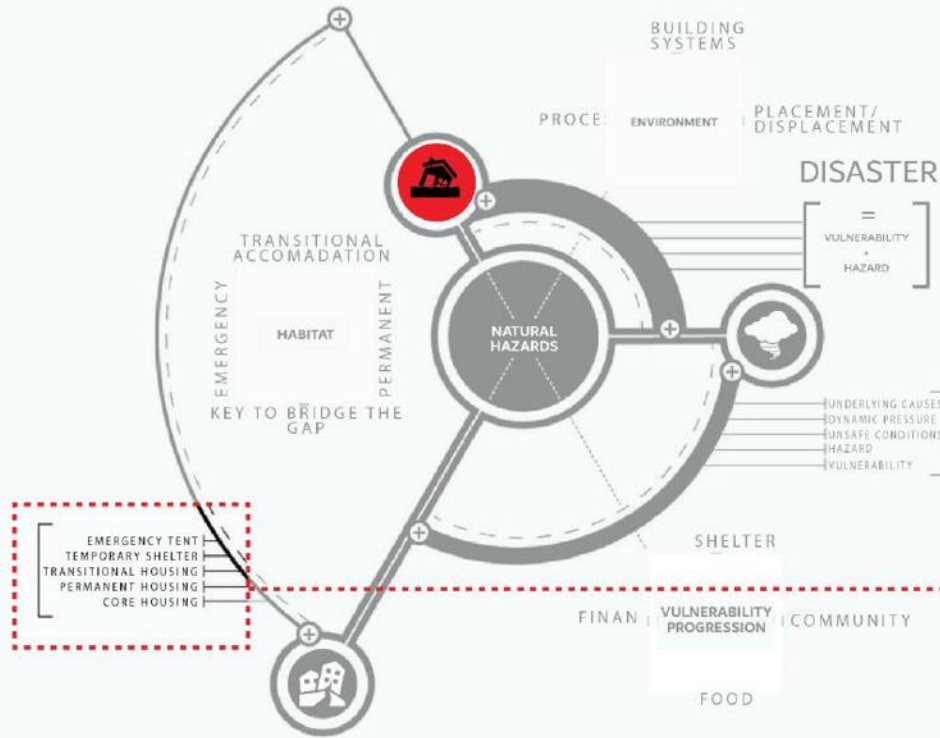
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A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

(06)



KEY WORD	REFERENCE	DEFINITION
HOUSING TYPOLOGY	LEUPEN ET AL 2011	Classification of housing based on physical, functional, and cultural characteristics.
SHELTER	JOHNSON 2007	A structure that provides protection from the environment, critical in disaster recovery.
TRANSITIONAL SHELTER	SHELTER CENTRE 2012	Temporary housing solution used in the recovery phase of a disaster, designed to be upgradable and adaptable.
CORE HOUSING	KUMAR 2010	A basic, minimal housing unit provided as a starting point for future improvements.

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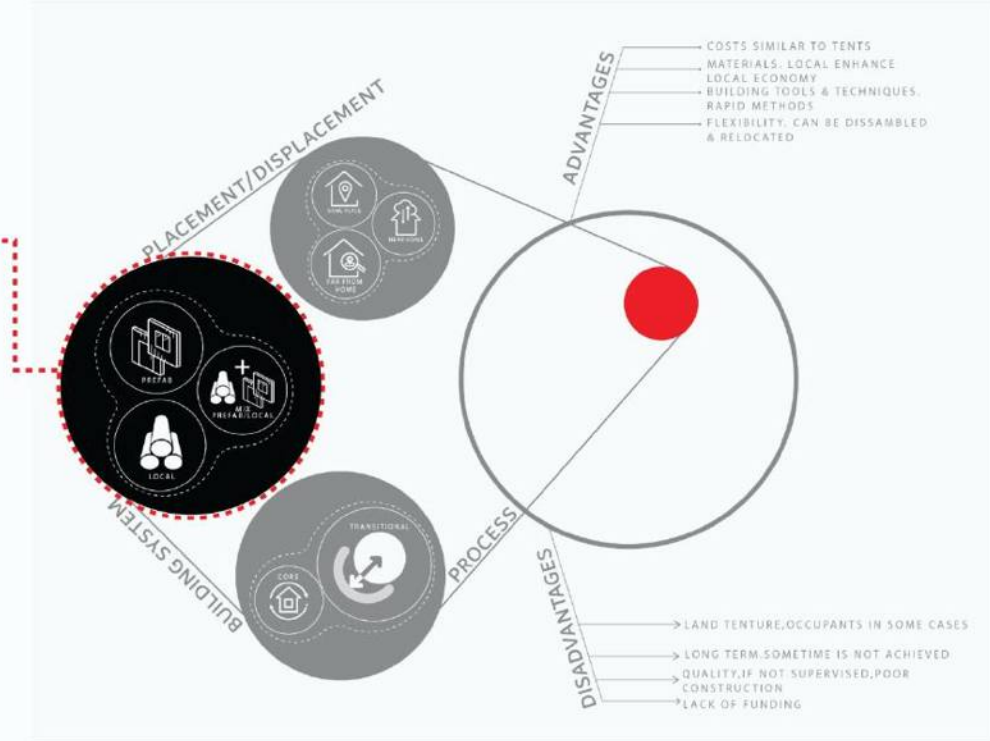
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UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

(06)

KEY WORD	REFERENCE	DEFINITION
RECOVERY PROCESS	GFDRR KNOWLEDGE NOTES: EARTHQUAKE RECONSTRUCTION - WORLD 2011	Stages following a disaster, involving emergency response, damage assessment, and reconstruction.
PROTOTYPE STRUCTURE	JOHNSON 2002	A prefabricated model or example of a shelter design used to test and refine before mass production.
RESILIENCE	SHAH ET AL 2018	The ability of a community or society to resist, absorb, and recover from the effects of a disaster.



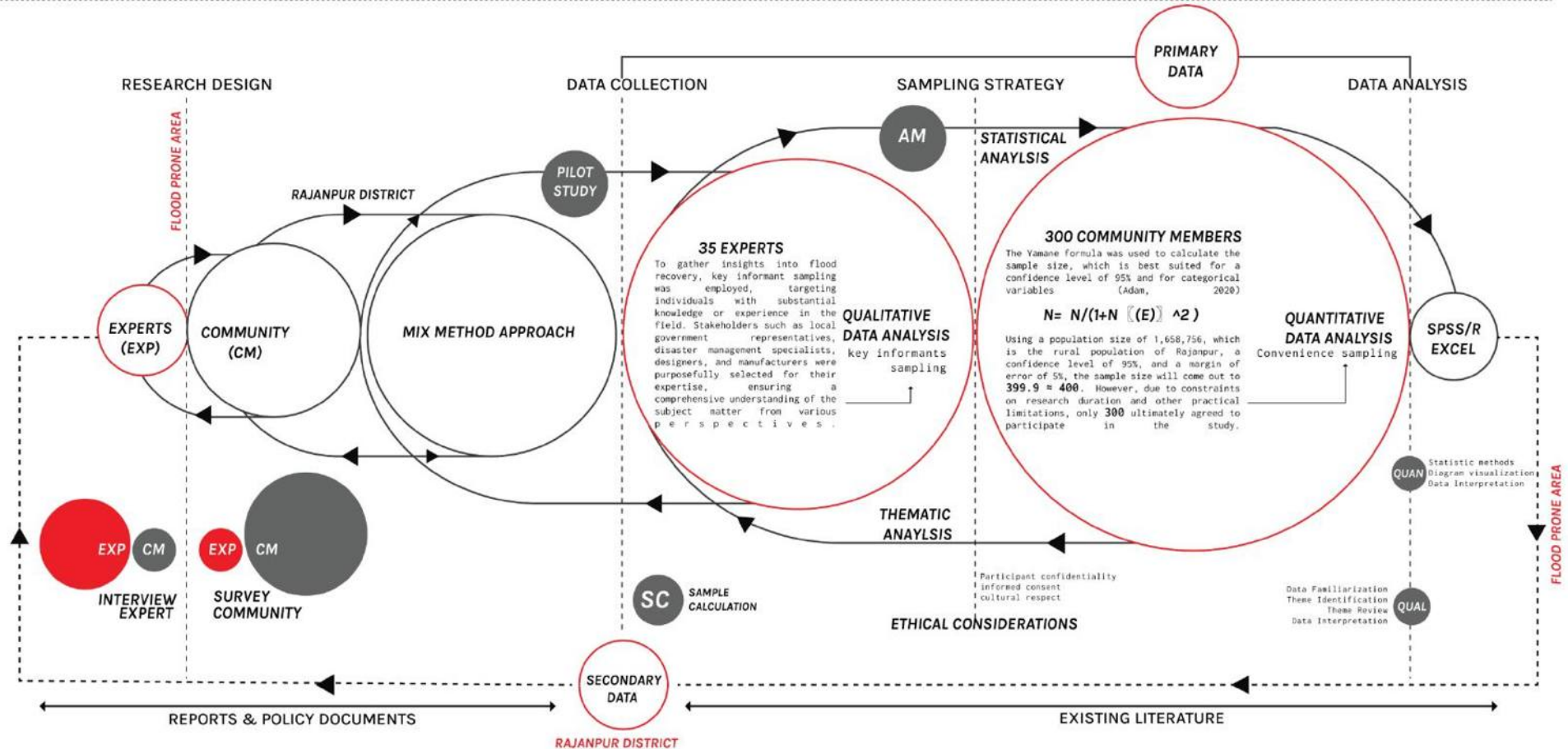
METHODOLOGY

- Research Design
- Conceptual Framework
- Indicators
- Associate Questions

UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

(07)

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN



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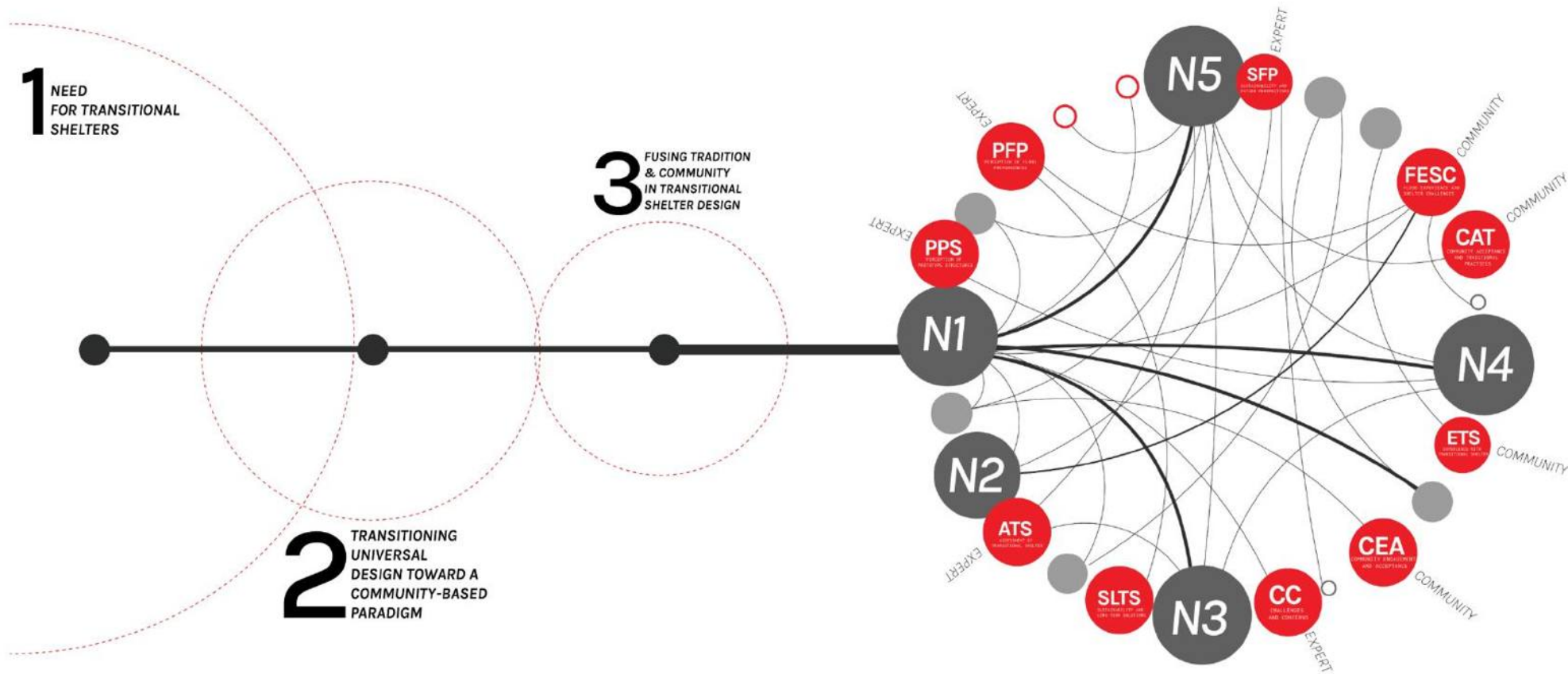
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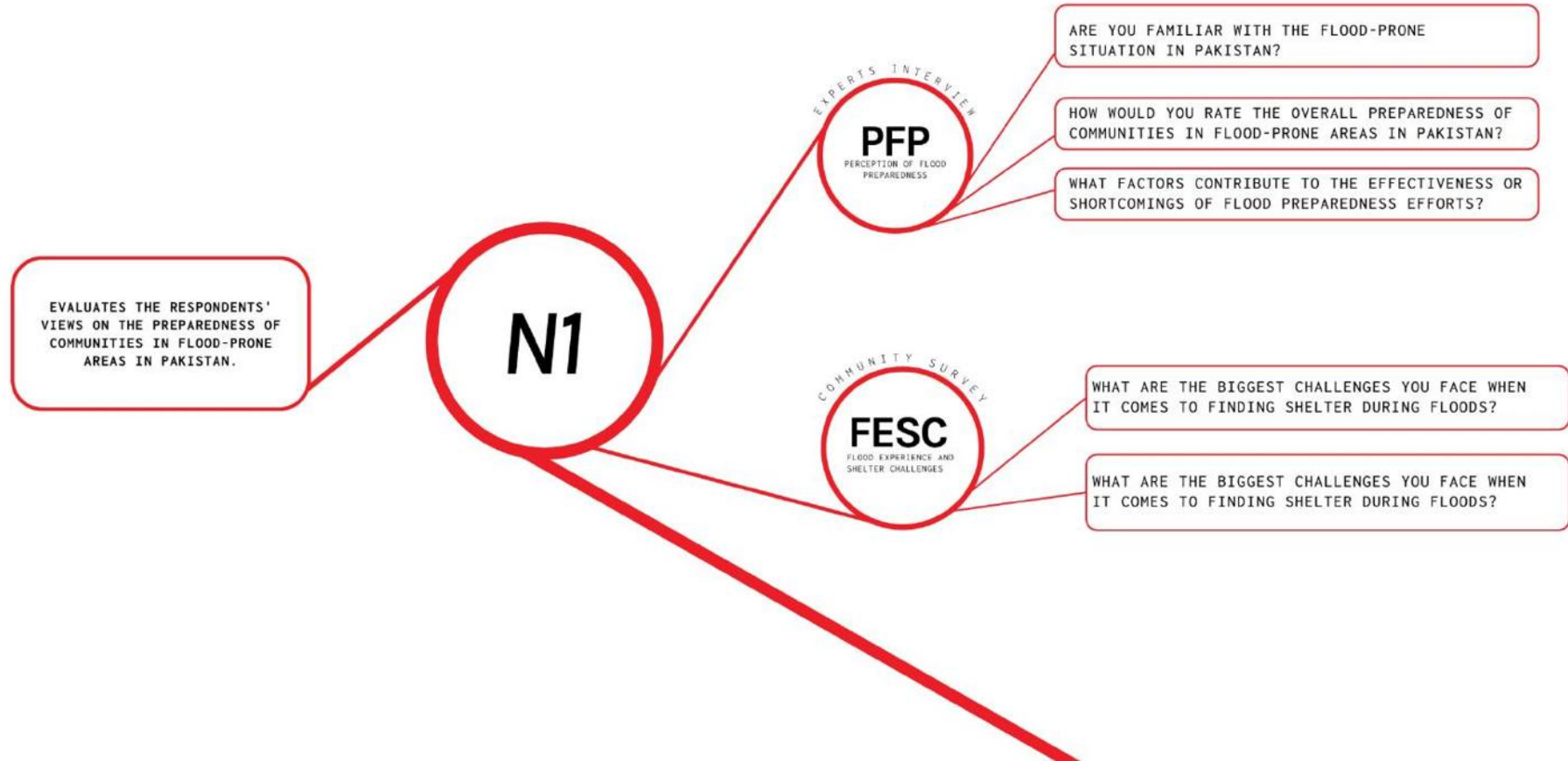
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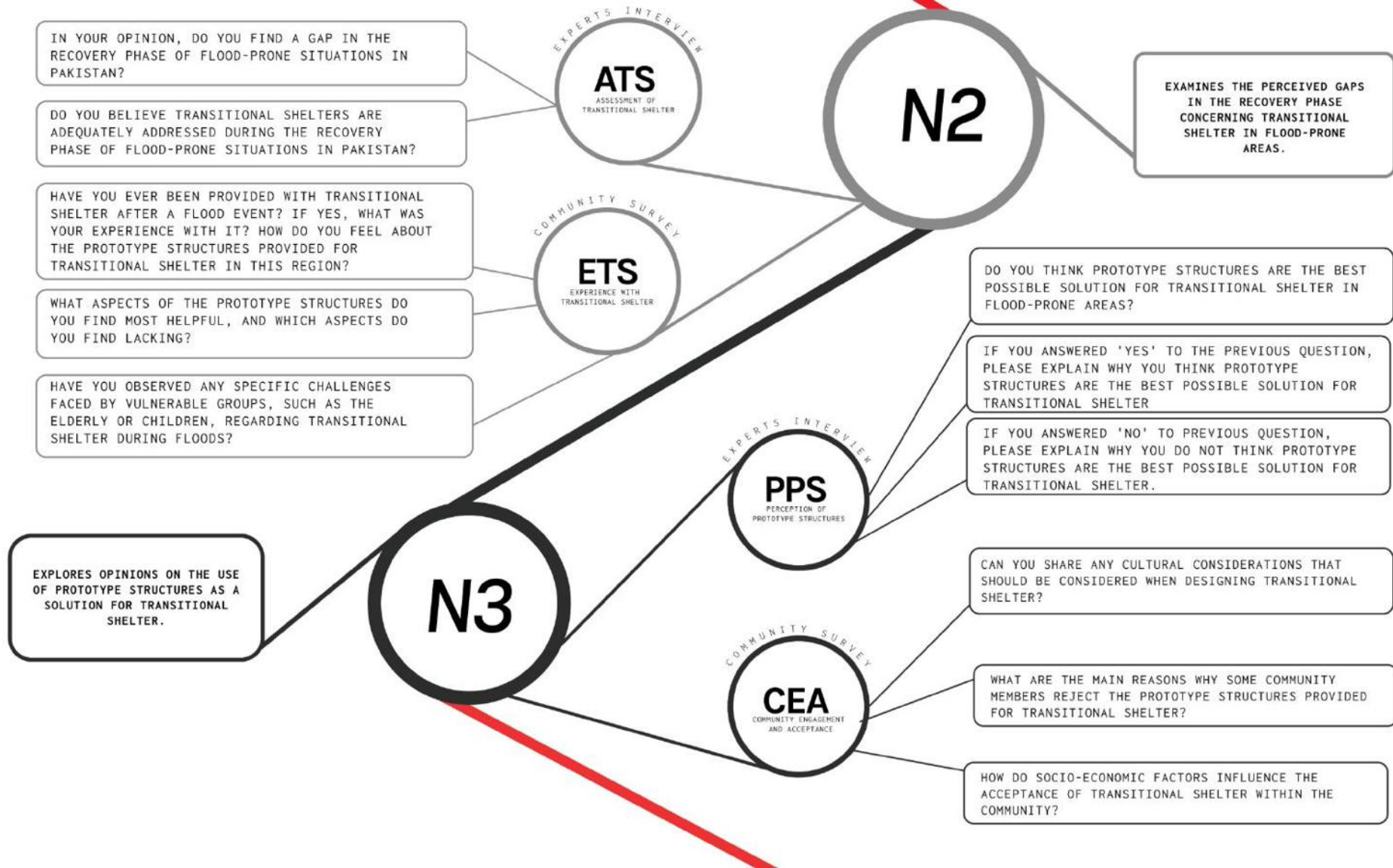
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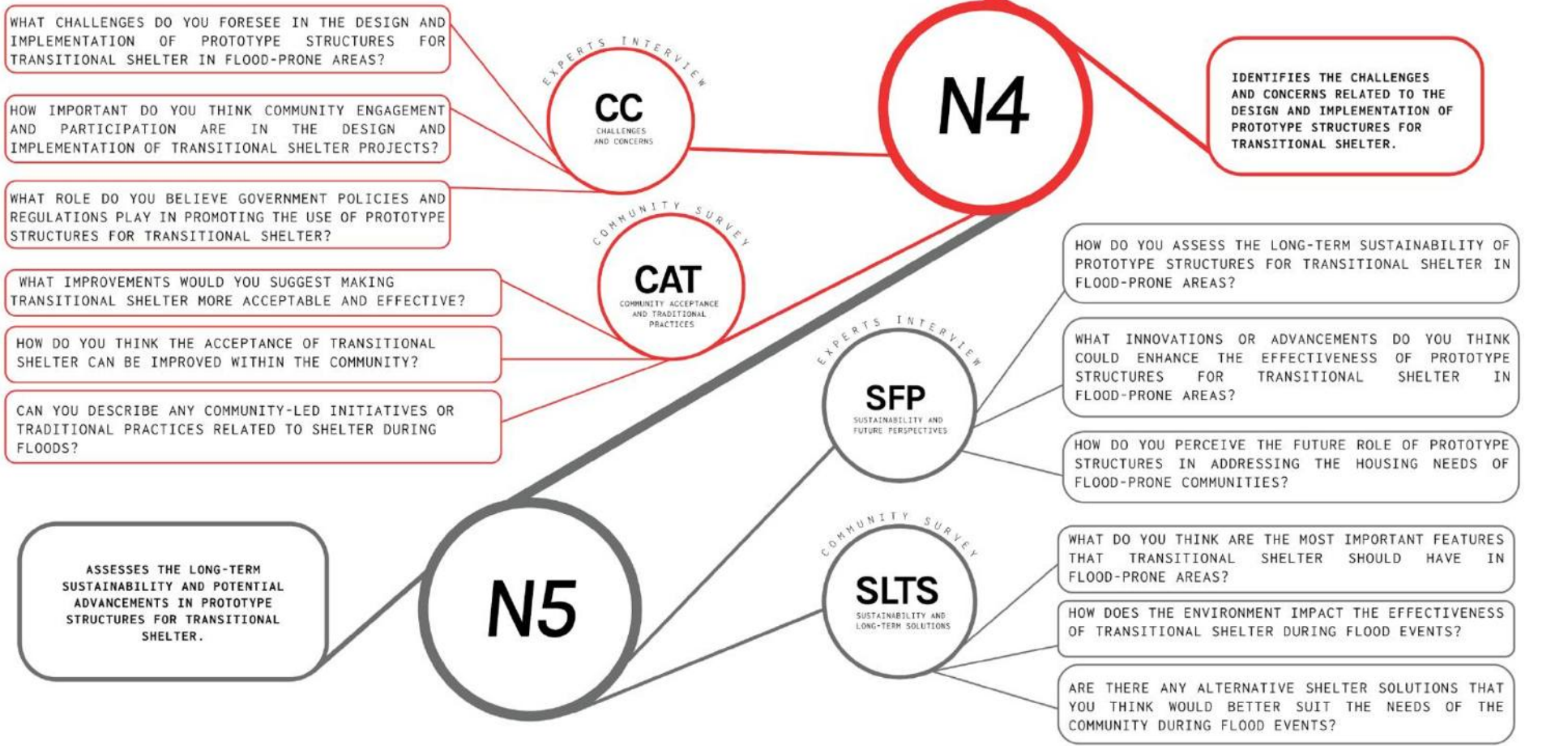
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DATA FINDINGS

- Expert Demographics
- Community Demographics
- Expert Opinion
- Community Opinion

UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

(10)

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

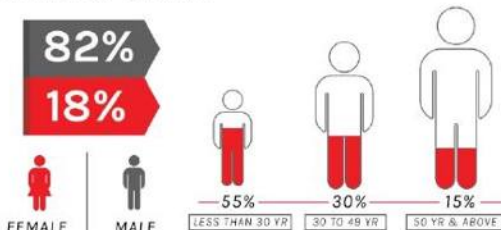
COMMUNITY OPINION

DEMOGRAPHICS

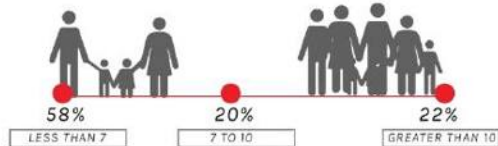
The study in flood-prone communities of Rajanpur District, involving 300 individuals across several villages, revealed key demographic insights. Males made up 78.6% of the population, highlighting gender disparities that need attention in disaster response. Age distribution showed that 55.3% were under 30, 29.7% were between 30 and 49, and 15% were 50 or older, indicating the need to address age-specific vulnerabilities.

Farming was the main livelihood, with 82% of participants being farmers, emphasizing the importance of agricultural resilience in flood recovery. Family sizes varied, with 58.3% having fewer than 7 members, 20.3% with 7 to 10 members, and 21.3% with more than 10 members, suggesting the need for adaptable shelter designs.

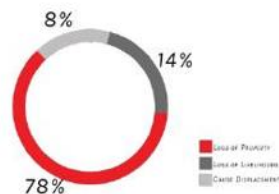
GENDER & AGE



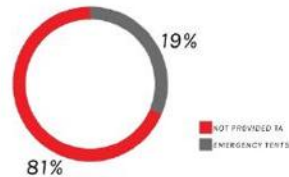
FAMILY SIZE



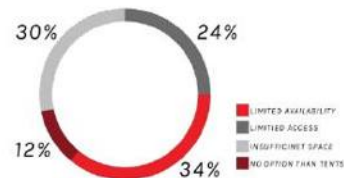
FLOOD EXPERIENCE



TRANSITIONAL EXPERIENCE



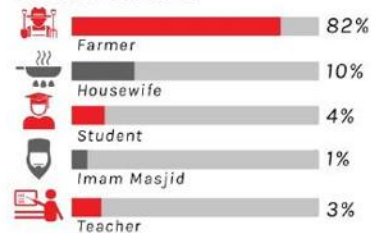
FINDING SHELTER



LOCATION



OCCUPATION



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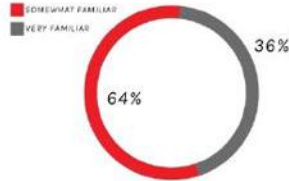
EXPERT OPINION

DEMOGRAPHICS

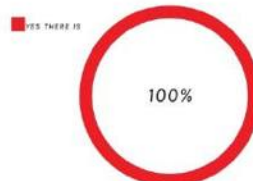
A total of 35 experts from various fields involved in flood relief efforts were consulted through a structured questionnaire. The demographic profile shows that 71.4% were male, and 28.6% were female. Most experts had 3 to 5 years of experience (45.7%), with job titles spanning NGOs, government institutions, architects, the army, researchers, and manufacturers.

About 34.3% had direct connections with flood events, while 65.7% had indirect connections. The participants' diverse age groups provided a range of perspectives, with the majority being young professionals with direct flood experience, offering valuable insights into transitional shelter and housing in flood-prone areas.

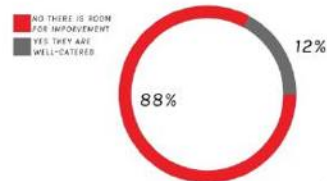
FLOOD-PRONE FAMILIARITY



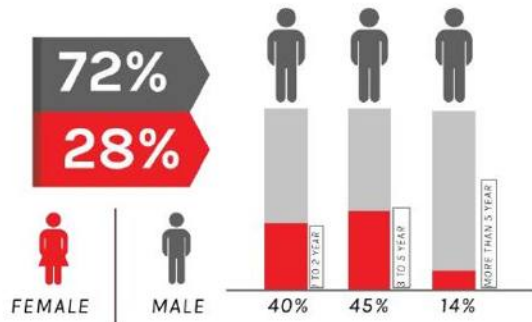
RECOVERY PHASE GAP



FOCUS TOWARDS TRANSITIONAL SHELTER



GENDER & EXPERIENCE



PROFESSION

NGO's (National & International)	25%
Govt. Institutions (NDMA, PDMA, CDPM, PDA)	15%
Architects	17%
Military Forces	9%
Reseachers	14%
Manufactures	20%

CONNECTION WITH FLOOD



SOURCE: MULTIPLE SOURCES

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UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

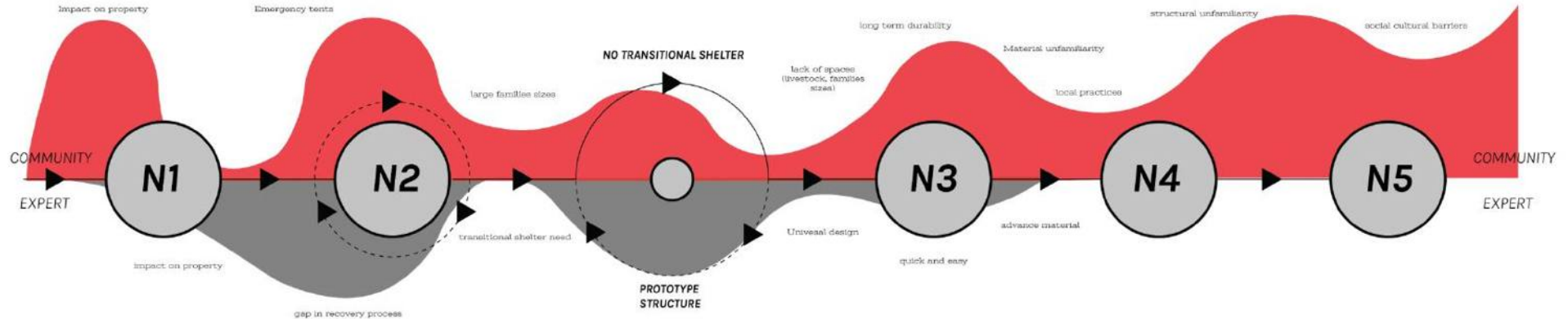
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A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

NEED FOR TRANSITIONAL SHELTERS

FUSING TRADITION AND COMMUNITY IN TRANSITIONAL SHELTER DESIGN

TRANSITIONING UNIVERSAL DESIGN TOWARD A COMMUNITY-BASED PARADIGM



SUMMARY

Implementing transitional shelters in the flood-prone Rajanpur District of Pakistan reveals a significant gap in disaster recovery efforts. Community involvement in shelter design and implementation is crucial to ensure cultural compatibility and practical usability.

While prototype structures offer innovative solutions, they face skepticism due to concerns about long-term sustainability and suitability for local needs. An integrated approach that blends traditional practices with modern innovations can create effective, resilient, and accepted shelter solutions for affected communities.

PROBLEM

Rajanpur faces a critical gap in shelter solutions. This research proposes merging local architecture with modern engineering for resilient shelters.

Shelter prototypes in Rajanpur fail to meet community needs. This study aims to design culturally suitable, resilient shelters.

OBJECTIVES

Analyze shelter challenges in flood-prone areas.

Gather expert insights to develop resilient, acceptable shelter solutions.

Understand community rejection of prototypes to improve future shelter designs.

FINDINGS

There is a critical gap in the provision of transitional shelters during the recovery phase in flood-prone areas of Pakistan, highlighting an urgent need for improved disaster response strategies.

The success of transitional shelter projects heavily depends on active community engagement, as it ensures that shelters meet the needs and preferences of the affected populations.

While prototype shelters offer potential, there is significant skepticism regarding their suitability, particularly concerning cultural compatibility, structural safety, and sustainability, indicating the need for further refinement and adaptation.

Flooding has a profound and recurring impact on communities, resulting in substantial property damage, disrupted livelihoods, and displacement, which underscores the necessity for resilient and effective shelter solutions.

There are significant doubts about the long-term sustainability of current prototype shelters, particularly regarding material durability and maintenance, suggesting that existing solutions may not be viable for extended use.

Effective shelter designs must integrate local cultural and traditional practices to ensure acceptance and usability, particularly in accommodating large families, livestock, and traditional building methods.

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RESULTS

1.GAP IN TRANSITIONAL SHELTER PROVISION

Critical shortfall in the provision of transitional shelters during the recovery phase in flood-prone areas, necessitating urgent improvements in disaster response strategies.

2.IMPORTANCE OF COMMUNITY ENGAGEMENT

Active engagement of the community in the design and implementation of transitional shelters is crucial for the success of these projects, ensuring that they meet the specific needs of the affected populations.

3.SKEPTICISM TOWARDS PROTOTYPE STRUCTURES

There is notable skepticism regarding the effectiveness of prototype structures, with concerns related to cultural compatibility, structural safety, and long-term sustainability. This highlights the need for further development and adaptation of these shelters.

4.SIGNIFICANT IMPACT OF FLOODING ON LIVELIHOODS

The communities surveyed have experienced severe impacts from flooding, including widespread property damage, disrupted livelihoods, and displacement, demonstrating the chronic nature of flooding in these areas.

5.CONCERNS ABOUT LONG-TERM SUSTAINABILITY

There are significant concerns about the long-term sustainability of current prototype shelters, particularly regarding their material durability and the need for ongoing maintenance, which may limit their viability as long-term solutions.

6.CULTURAL COMPATIBILITY IN SHELTER DESIGN

The success of transitional shelters depends on their ability to incorporate local cultural and traditional practices. Shelters must accommodate large families, livestock, and traditional building methods to be effective and accepted by the community.

7.GOVERNMENT SUPPORT & FUNDING

The research highlights the critical role of government support and funding in the success of transitional shelter projects. Adequate resources and streamlined processes are essential to overcoming challenges in the design, implementation, and sustainability of these shelters.

"TRANSITIONAL SHELTERS: A VITAL NEED"

- Analysis
- Discussion
- Conclusion

DATA FINDINGS

NEED FOR TRANSITIONAL SHELTERS

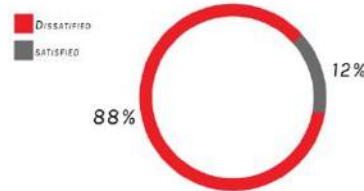
Experts highlight the crucial role of transitional shelters, particularly prototype models, in disaster relief and recovery. Despite their importance, 88% of experts express dissatisfaction with current provisions, citing significant gaps in shelter availability during the recovery phase. Communities view these shelters as essential for immediate relief but stress the need for them to be integrated into a broader recovery strategy.

PHASES OF SHELTER PROCESS

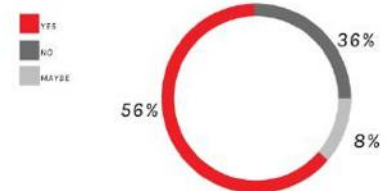
CURRENT TWO PHASE PROCESS IN RAJANPUR

Global examples, like those from Haiti and the Philippines, show the effectiveness of transitional shelters in managing disaster impacts. In Rajanpur, Pakistan, the reliance on tents prolongs the emergency phase, bypassing the intended transitional stage, which underscores the need to revisit and reinforce the three-phase shelter process to enhance disaster recovery.

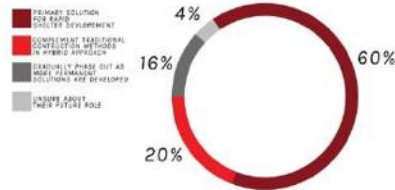
EXPERT SATISFACTION WITH TRANSITIONAL SHELTER PROVISION (EXPERTS)



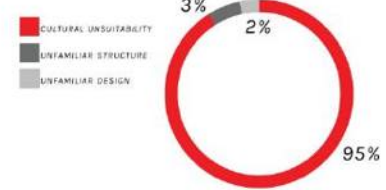
PROTOTYPE STRUCTURES THE BEST SOLUTION (EXPERTS)



FUTURE OF PROTOTYPE STRUCTURES (EXPERTS)

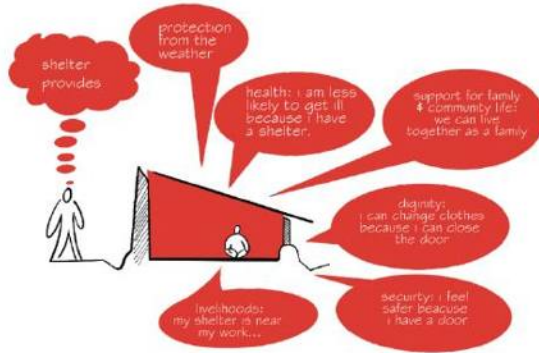


FEELINGS ABOUT PROTOTYPE STRUCTURES (COMMUNITY)



ANALYSIS

1 CRUCIAL ROLE OF SHELTER



SOURCE: SHELTER PROJECTS ESSENTIALS (2023)

Transitional shelter provides a habitable covered living space and a secure, healthy living environment, with privacy and dignity, to those within it, during the period between a conflict or natural disaster and the achievement of a durable shelter solution

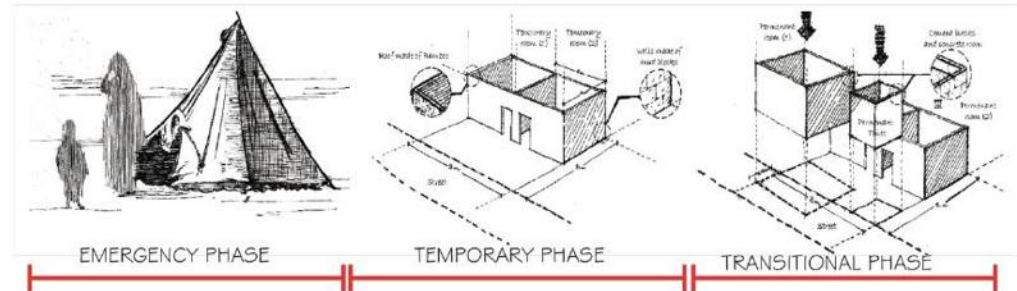
Transitional shelters should be durable, easy to upgrade or relocate, and built quickly using methods suited to the local environment. These shelters are especially helpful for displaced and vulnerable populations, aiding their recovery, though more research is needed on tailored strategies for different groups.

2 DURABILITY AND FLEXIBILITY OF TRANSITIONAL SHELTERS

Transitional shelter solutions should be durable enough to last the entire transition period until reconstruction is complete and should be able to upgrade or relocate to different sites.

It should be simple techniques and rapid construction methods appropriate to the environment and community with which it is being implemented. e 'rapid reconstruction' may be the most efficient way to support recovery for most of an affected population, transitional shelter may be particularly beneficial to displaced populations (enabling them to return home or at least establish some degree of normalcy in a temporary location), vulnerable households (who may need additional support/time to engage in 'rapid reconstruction' programs) or other specific interest groups.

This disaggregation of shelter strategies for different groups within the affected population is not common within the shelter sector and would benefit from further research.



SOURCE: RECOVERY PHASE CONSTRUCTION

3 CHALLENGES OF THE THREE-PHASE RECONSTRUCTION STRATEGY

While this strategy seems promising in theory, it faces significant challenges in practice. The durable solutions—repatriation, resettlement, and local integration—often remain unattainable, leaving people stuck in emergency shelters for years or even decades.

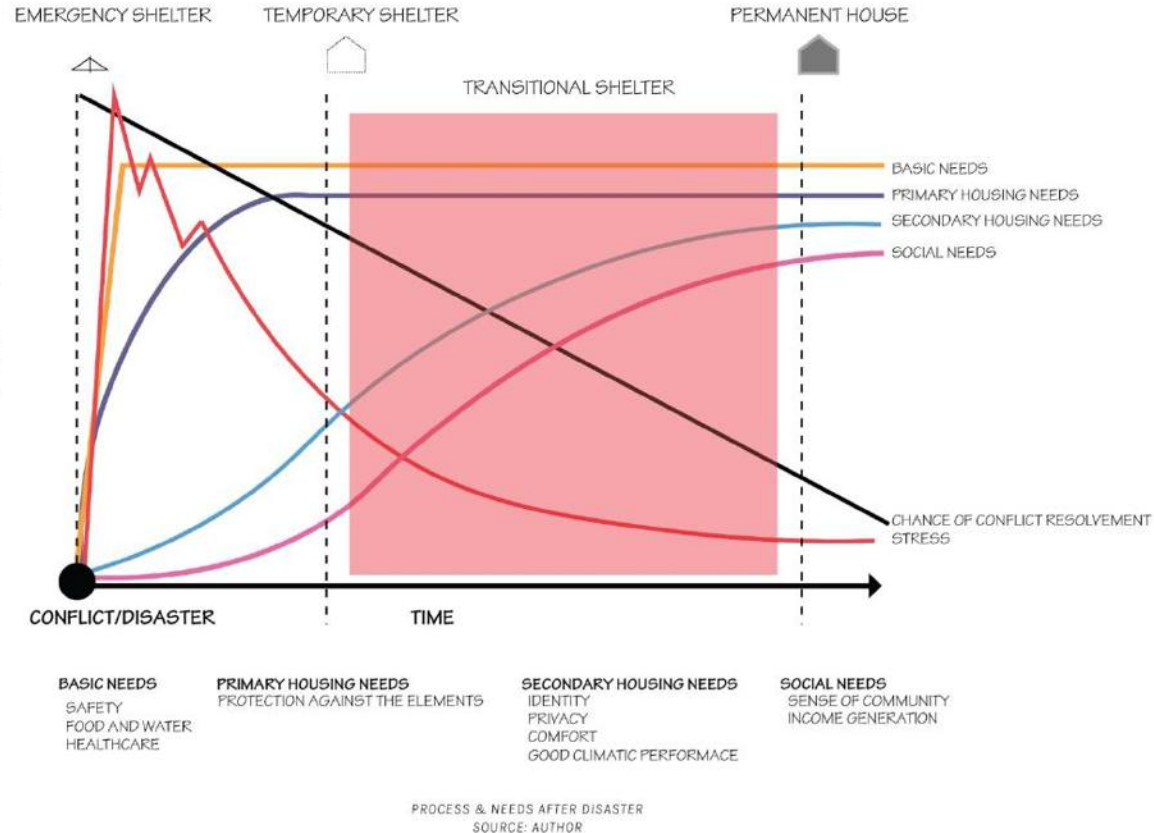
These shelters, designed for temporary use, provide only basic protection against the elements, lacking qualities needed for long-term living, such as privacy and adequate thermal insulation.

Meeting these needs requires constructing permanent housing, but the necessary funds are often unavailable. As a result, many remain in substandard conditions, with only a few able to replace makeshift shelters with salvaged or purchased materials, which are often too expensive or restricted by UNHCR regulations.

4 NEED FOR INCREMENTAL APPROACH

Implementing an incremental approach with transitional shelters in Rajanpur is crucial for effective disaster recovery.

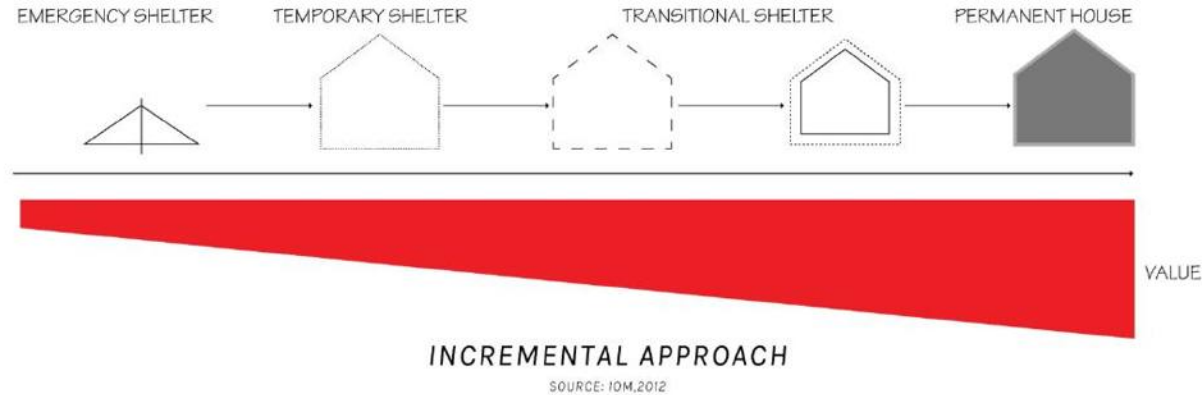
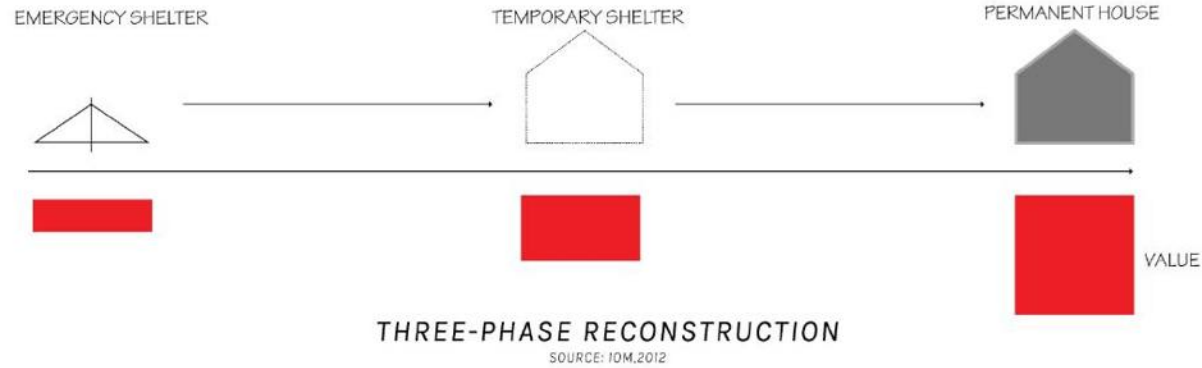
This strategy not only fills the gap left by the current two-phase system but also supports the community's long-term needs. By incorporating transitional shelters into the recovery process, the community gains resilience, cultural alignment, and sustainable housing solutions that can evolve with their changing circumstances.



UNDERSTANDING PROTOTYPE STRUCTURES FOR TRANSITIONAL SHELTER FOR RURAL FLOODING

A CASE STUDY OF RAJANPUR DISTRICT, PAKISTAN

(14)



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CONCLUSION



THE MISCONCEPTION OF TRANSITIONAL SHELTERS AS PRODUCTS

Davis's concept that shelter is a 'process' rather than a 'product' has been echoed by Corsellis and Vitale, advocating for supporting families towards a durable solution. However, the current focus in the shelter sector often prioritizes the design and specification of transitional shelters as products, neglecting the longer-term reconstruction process. This approach risks leaving families vulnerable after the transitional phase ends, especially when agencies withdraw support.

THE INCREMENTAL PROCESS OF TRANSITIONAL SHELTER

Transitional shelter is often misunderstood as a one-time product instead of an incremental process that starts with emergency relief and evolves into more permanent housing. Prefabricated shelters, though sometimes used as transitional solutions, are costly and do not engage the community, making them less suitable, especially in rural areas.

ADDRESSING LAND TENURE AND LONG-TERM STABILITY

For populations like farmers in transitional living conditions, who often lack land tenure, transitional shelters can offer temporary stability while land issues are resolved. These shelters support the ongoing development of comprehensive recovery strategies, enabling affected populations to gradually rebuild their lives and homes. Transitional shelters should be part of a broader, integrated strategy that starts from the initial emergency phase and runs parallel to permanent reconstruction efforts, allowing for upgrades, reuse, or resale as needed.

TRANSITIONAL SHELTER AS A PROCESS NOT A PRODUCT

"UNIVERSAL DESIGN FOR ALL"

- Analysis
- Discussion
- Conclusion

DATA FINDINGS

EXPERT EMPHASIS ON RAPID ASSEMBLY & TECHNICAL SOUNDNESS

Experts emphasize the need for practical design guidelines that prioritize quick assembly and technical soundness in transitional shelters. Their focus is on creating shelters that can be rapidly deployed with universal design principles, ensuring they are easy to assemble, durable, and adaptable to various site conditions. The goal is to streamline the process for immediate relief while maintaining safety standards.

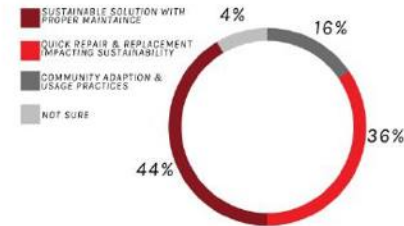
COMMUNITY FOCUS ON FLEXIBILITY & CULTURAL RELEVANCE

In contrast, the community's perspective is more focused on flexibility and cultural relevance. They stress the importance of designs that accommodate different family sizes, address privacy concerns, and support cultural practices. Community members also highlight the need for shelters that provide space for livestock, which are essential to their livelihoods. They advocate for designs that respect vernacular architecture and materials to enhance cultural acceptance.

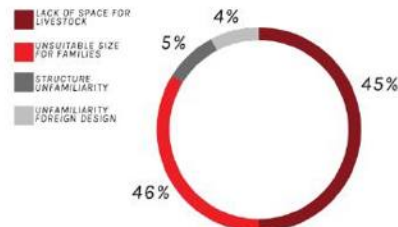
INTEGRATING EXPERT & COMMUNITY PERSPECTIVES FOR HOLISTIC SOLUTIONS

While experts prioritize technical efficiency, the community seeks adaptable designs that meet their diverse and cultural needs. Integrating both perspectives can lead to shelters that are both technically robust and socially responsive, improving disaster recovery outcomes.

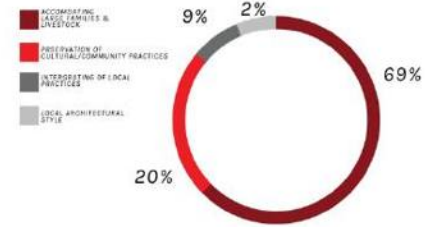
LONG TERM SUSTAINABILITY OF PROTOTYPE STRUCTURES (EXPERT)



REASONS FOR REJECTION (COMMUNITY)



CULTURAL CONSIDERATIONS (COMMUNITY)



ANALYSIS

1 UNHCR MINIMUM STANDARDS FOR TRANSITIONAL SHELTERS

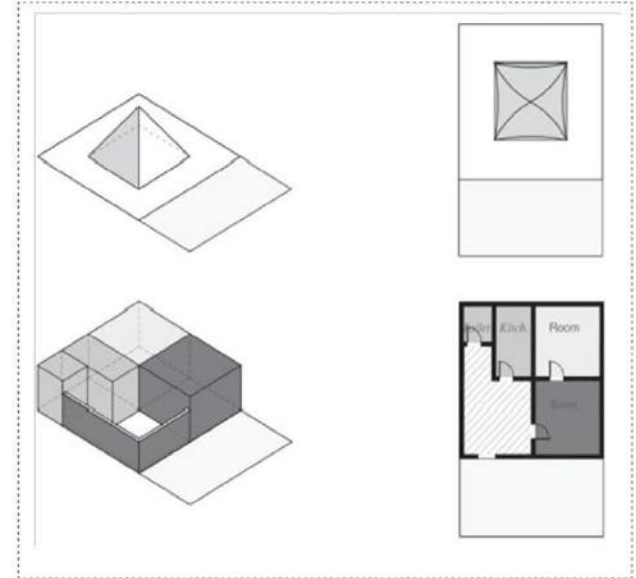
The UNHCR outlines minimum standards for floor space in warm climates to ensure adequate living conditions for displaced populations. These standards specify a minimum of 37.6 square feet of covered living space per person, excluding cooking facilities or kitchens. The height of shelters should be at least 7 feet at the lowest point, with a greater height preferred to enhance air circulation and ventilation. In warm climates, cultural habits often dictate that activities like cooking occur outside, thus shaded external spaces adjacent to shelters are recommended for such activities.

2 EMERGENCY PHASE: INITIAL SHELTER SETUP

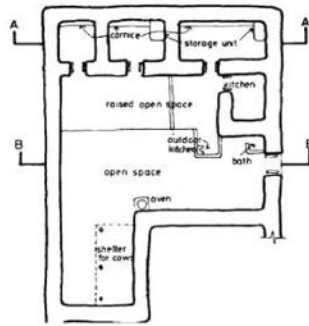
During the emergency phase, displaced people are provided with 26 x 26 ft plots upon arrival at the camp, where they set up tents as temporary shelters. Government and NGOs distribute these tents, but the internally displaced persons (IDPs) are responsible for setting them up. These tents typically serve as living spaces for 1 to 2 years until the floodwaters recede and reconstruction of permanent houses begins.

3 TRANSITIONAL PHASE: ENHANCING LIVING CONDITIONS

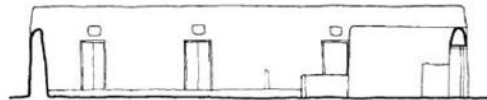
In the transitional phase, displaced people expand their initial one-room dwellings by adding kitchens and bathrooms to improve living conditions. This phase involves greater community participation, with individuals enhancing their living spaces using more durable and vernacular materials. The use of local materials ensures cultural relevance and sustainability, reflecting the community's needs and preferences.



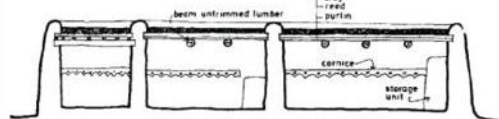
4 RURAL DWELLINGS OF PUNJAB



The typical rural dwelling in Kotkaramat, South Punjab, provides an insightful case study into the nature of permanent dwellings in rural Punjab, reflecting a lifestyle that is inherently transitional. These homes are characterized by a central raised open space surrounded by essential functional areas, embodying simplicity and practicality necessary for the transient lifestyle of many rural inhabitants.



SECTION BB

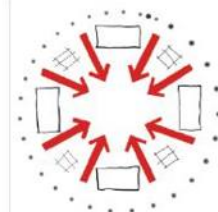


SECTION AA

SOURCE: TRADITIONAL FORMS OF RURAL HABITAT, A STUDY FOR UNESCO, 1977 ISSJ VOL XXX 1978 NO. 3.

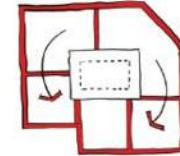
The layout includes two primary rooms for living and sleeping, storage units, a grain store, a chicken cage, and an area for pitchers. Traditional materials like clay, reed, and untrimmed lumber are used, highlighting reliance on locally available resources and ensuring cultural relevance and sustainability.

This design reflects the "kami" system, where individuals work on land they do not own, necessitating flexibility and mobility.



CENTRAL COMMUNAL SPACE FOR COMMUNITY COHESION

Emphasizes the importance of a communal area that facilitates daily activities and social interaction, crucial for building community ties.



MULTIPURPOSE ROOMS FOR ADAPTABILITY

Designs rooms that can serve various functions based on immediate needs, ensuring the adaptability of the living space.



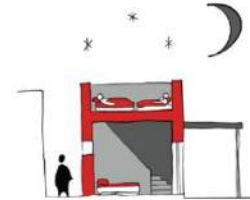
UPGRADABLE BASIC STRUCTURES

Ensures that the basic structure of shelters can be easily upgraded over time to improve living conditions gradually.



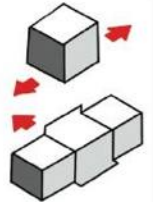
CULTURAL RELEVANCE

Utilizes materials that are cost-effective, sustainable, and culturally appropriate, promoting local building practices.



INTEGRATED STORAGE & LIVESTOCK

Integrates essential spaces for storage and livestock within the dwelling to support the agricultural practices of the community.



SMOOTH TRANSITION FROM EMERGENCY RELIEF

Supports a seamless transition in shelter use from temporary emergency relief to more stable, permanent living conditions, accommodating the unique needs and cultural practices of the community.

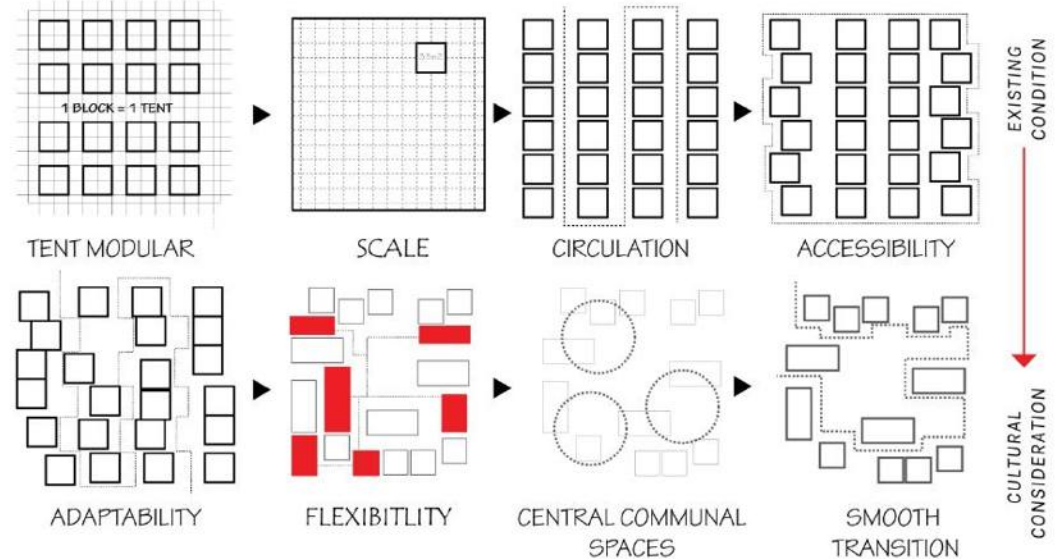
CONCLUSION

PHASED APPROACH TO SHELTER DESIGN: TRANSFORMING EMERGENCY SHELTERS INTO SUSTAINABLE COMMUNITY HOUSING

The strategy begins with the Emergency Phase, deploying prefabricated units quickly to provide basic protection and privacy for displaced individuals. In the Temporary Phase, the community builds one-room dwellings with temporary materials, enhancing adaptability and a sense of ownership.

The Transitional Phase expands these structures to include kitchens and bathrooms, using durable materials for sustainable and culturally relevant shelters. Finally, the Community Phase develops permanent housing that integrates with the community's broader needs, such as privacy, sanitation, and communal spaces.

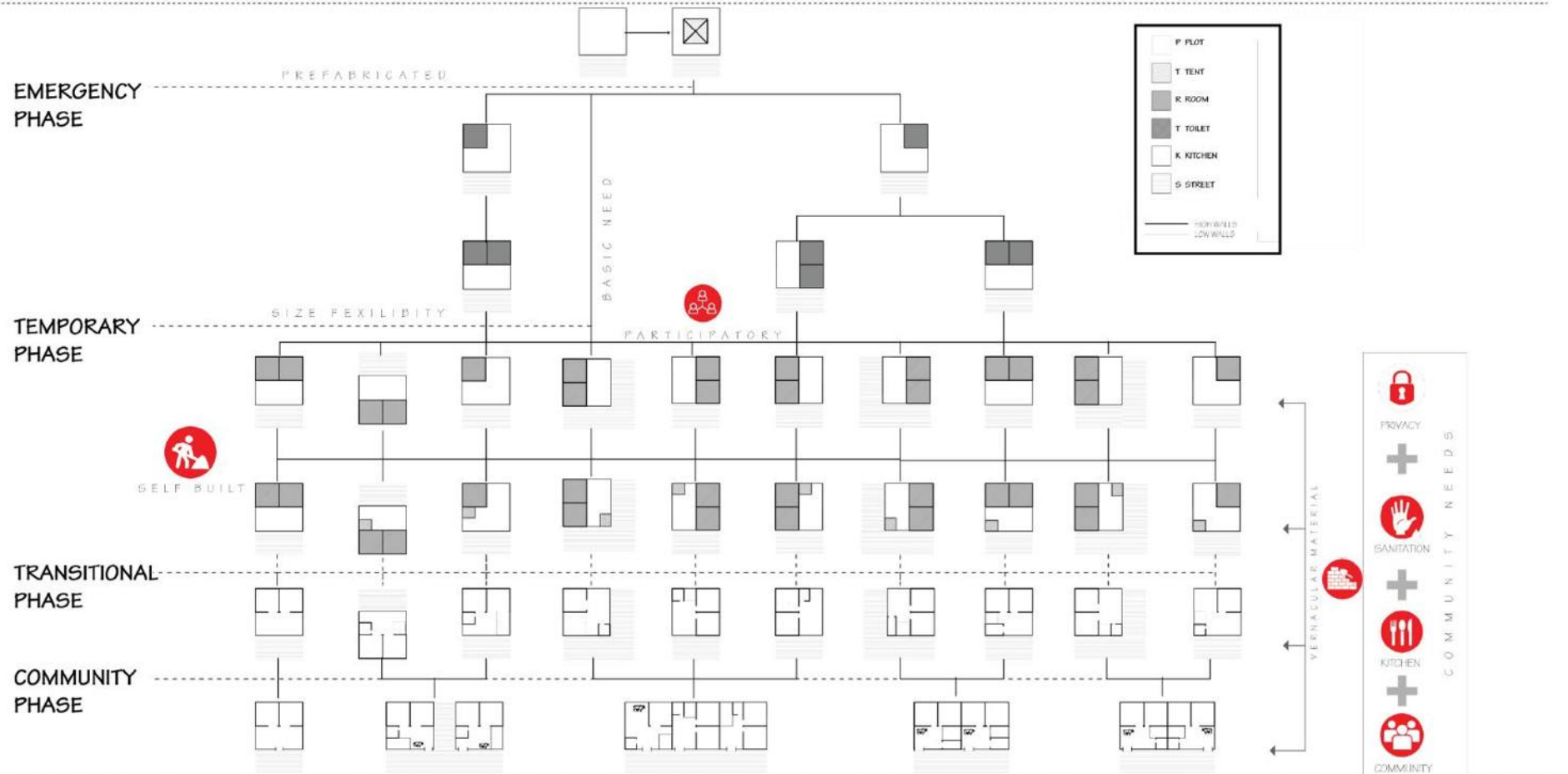
This approach emphasizes privacy, sanitation, and communal areas throughout, ensuring shelters are technically efficient and socially responsive. By involving the community at each stage, the strategy supports long-term recovery and transforms shelters into cohesive units that reflect the community's unique needs and cultural practices.



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"COMMUNITY-CENTERED, CULTURALLY AWARE"

- Analysis
- Discussion
- Conclusion

DATA FINDINGS

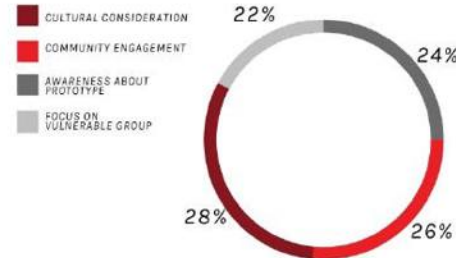
COMMUNITY ENGAGEMENT AND CULTURAL CONSIDERATIONS IN TRANSITIONAL HOUSING

Both experts and community members agree on the importance of community engagement in the success of transitional housing projects. Involving the community in decision-making fosters ownership, acceptance, and overall well-being, leading to better recovery outcomes. This engagement ensures that shelters meet specific needs and cultural practices, and remain adaptable over time.

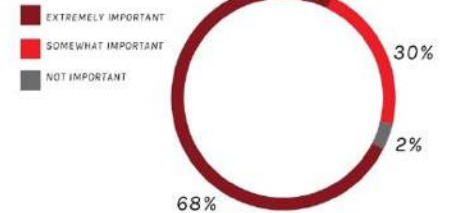
Cultural considerations and sustainability are also highlighted as key factors. Both groups emphasize the need for shelters to be adaptable for long-term use, integrating sustainable practices and vernacular methods to maintain cultural identity. This approach not only addresses immediate needs but also ensures the durability and relevance of housing solutions in the future.

The shared understanding between experts and community members underscores the need for a holistic approach to shelter design. By valuing community input and integrating sustainable practices, it's possible to create shelters that are both effective in the short term and adaptable for the long term.

ACCEPTANCE IMPROVEMENT



IMPORTANCE OF COMMUNITY ENGAGEMENT



ANAYLSIS

1 ROLE OF COMMUNITY PARTICIPATION IN VERNACULAR ARCHITECTURE



Community participation is fundamental in vernacular architecture, rooted in engagement and collective efforts. Unlike modern practices, vernacular traditions involve everyone in the design and construction, reflecting communal values and cultural identity.

2 SOCIO-CULTURAL INFLUENCE ON HOUSE FORMS



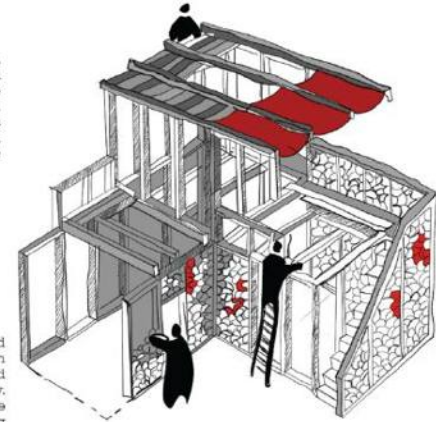
House designs in vernacular architecture are shaped by socio-cultural factors, linking physical needs with cultural and social values. Traditional practices highlight the importance of community-driven design, free from theoretical or aesthetic constraints.

3 COMMUNITY ENGAGEMENT IN DISASTER RECOVERY

In Rajanpur's flood recovery, community involvement and cultural considerations are crucial for the success of transitional housing. Engaging the community fosters ownership, acceptance, and resilience, ensuring shelters meet specific cultural needs.

4 SUSTAINABILITY AND VERNACULAR METHODS

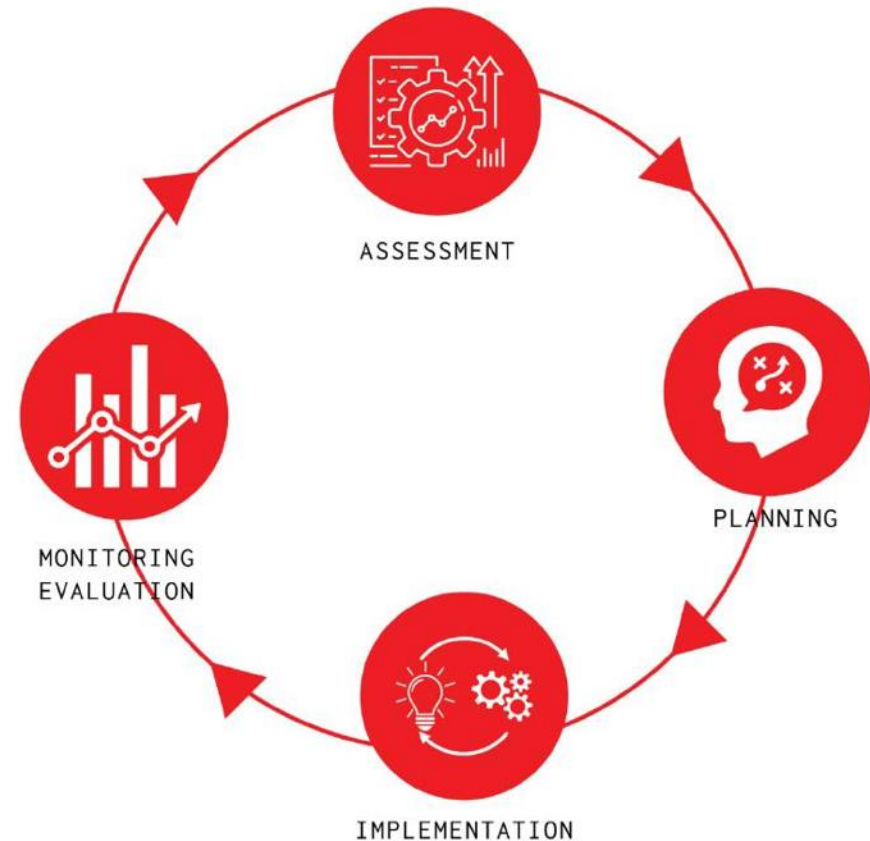
Incorporating vernacular methods and sustainable practices in shelter design enhances cultural relevance and environmental sustainability. Traditional techniques reduce reliance on imported materials, promoting resilience in disaster recovery.



CONCLUSION

COMMUNITY-CENTERED DISASTER RECOVERY: INTERNATIONAL GUIDELINES AND BEST PRACTICES

In accordance with international guidelines such as those outlined by the United Nations Office for Disaster Risk Reduction (UNDRR) and the Sphere Standards, this disaster recovery plan is structured around the key phases of assessment, planning, implementation, and evaluation. The plan emphasizes the importance of community involvement, respect for local customs, and the use of vernacular methods and materials to ensure that recovery efforts are sustainable, culturally appropriate, and effective in meeting the needs of the affected populations. By adhering to these principles, the plan seeks to build resilience and empower communities to recover from disasters with dignity and long-term success.



GUIDELINES FOR AFFECTED COMMUNITY

Involve the community from the start, ensuring inclusive participation from women, elders, craftspeople, and leaders. Recognize the roles of men and women, particularly women's contributions to childcare, cooking, and crafts. Assess local materials like mud, clay, straw, and wood, and investigate social structures for decision-making. Evaluate the impact on agriculture, livestock, and crafts, and identify recovery needs.

CULTURAL TOOLS & ACTIVITIES

Respect Local Customs and Traditions: Respect the cultural norms and beliefs of the community during assessments for better engagement.

Use Local Languages and Dialects: Facilitate communication with interpreters, ensuring culturally appropriate communication.

IMPLEMENTATION IN RAJANPUR

Rajanpur District's traditional practices strongly influence both construction methods and community tools. Mud-brick walls with clay plaster provide insulation, while Kacha mud walls and Sarai thatched roofs are cost-effective and quick to assemble, ideal for post-disaster shelters. Wattle fencing with mud plaster is commonly used for livestock enclosures, though durability against weather needs assessment.



ASSESSMENT

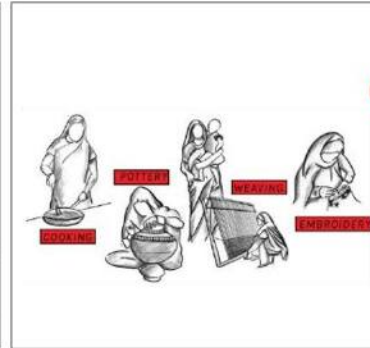
This phase involves evaluating the situation and engaging the affected community to identify their needs and ensure their involvement in the recovery process from the start.

DISASTER RECOVERY PHASE

TOOL



Panchayat (Local Governance): This social structure serves as a vital tool for managing the decision-making process. The Panchayat should be responsible for overseeing community involvement in every stage of the assessment and ensuring that cultural norms and gender-specific needs are addressed.



Pottery and Weaving Tools: In Rajanpur, pottery wheels and weaving looms are essential tools used by local artisans. These crafts should be recognized not only as economic activities but as cultural tools that shape the physical spaces in homes and workshops.



Siraiki Language: This tool ensures that all segments of the community, especially women and elders, can fully participate in discussions. It also helps maintain cultural integrity by keeping communication rooted in local traditions.



PLANNING

In this phase, community engagement is prioritized to incorporate their needs and preferences into the recovery plan.

GUIDELINES FOR AFFECTED COMMUNITY

Engage the community to ensure that their needs, knowledge, and skills are integrated into the design process. Create participatory design workshops where local leaders, artisans, women, farmers, and elders collaborate with planners and architects to design the built environment.

Women's input should be essential, especially regarding household spaces, cooking areas, childcare zones, and spaces for craft production.

Local materials and construction techniques must be the basis of design when feasible. Design should prioritize using locally available materials, incorporating these into modern designs when necessary to improve durability and resilience. For communities with specific traditional building techniques, ensure that these methods are used unless they pose significant risk.

The primary materials for construction in Rajanpur will be kachcha eent (mud-brick) and khaprail (thatch) due to their local availability and historical use. Gara plaster (mud mixed with lime or choona) will be applied to walls to prevent erosion and enhance durability against rain and wind. This traditional practice, known locally as lepna, has been used for generations to protect homes from the harsh climate.



Engage Mistris who have traditional knowledge and expertise in using local materials like kachcha eent (mud-brick) and chhajje (thatch roofing). Their skills in construction methods unique to Rajanpur, such as adobe brickmaking and mud plastering.

TOOL



Design spaces that reflect local cultural practices, such as communal chowks (courtyards) for family gatherings and rituals, while ensuring these are built on raised foundations to mitigate flood risk. Respect gender-specific needs by creating separate areas for men and women, such as zanana (women's) spaces for cooking, childcare, and crafts. This approach ensures that the built environment is aligned with social norms and local lifestyles.

Use traditional construction materials, such as kachcha eent (mud-brick) and chhajje (thatch), reinforced with local techniques like choona (lime) plastering to increase durability against weather impacts. The integration of these materials respects cultural norms while enhancing structural resilience. Engage Mistris (local craftsmen) to lead these efforts, ensuring that building techniques are both culturally appropriate and safe.

KACHCHA EENT & KHAPRAIL

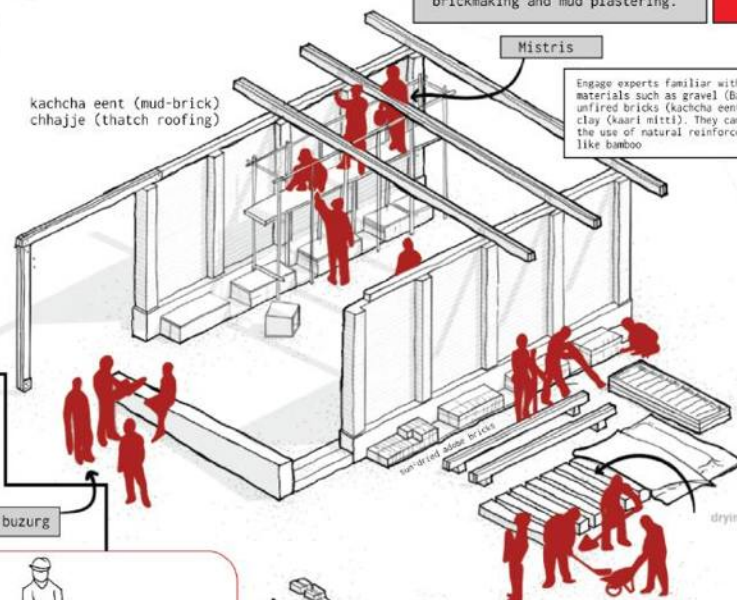
haathi lagana
(group lifting for construction tasks)

Involve local elders and Saangis (community storytellers) who are knowledgeable about the community's cultural practices, ensuring the design respects local customs, festivals, and gender roles. Their input is crucial for designing spaces that accommodate cultural activities and maintain social harmony.

kachcha eent (mud-brick)
chhajje (thatch roofing)

Mistris

Engage experts familiar with local materials such as gravel (Bajri), unfired bricks (kachcha eent), and clay (kaari mitti). They can guide the use of natural reinforcements like bamboo.



zila ke buzurg

IMPLEMENTATION IN RAJANPUR

Promote the use of pukka eent (sun-dried adobe bricks) by setting up temporary kilns in communities, empowering locals to produce bricks on-site.



pukka eent
(sun-dried adobe bricks)

Participatory construction
taamiri din

VERNACULAR METHODS & MATERIALS

GUIDELINES FOR AFFECTED COMMUNITY

"Involve the Community in Recovery Activities: Ensure community members are actively involved in the rebuilding efforts to enhance ownership and ensure the sustainability of the shelters.

Respect Cultural Practices During Implementation: Align construction schedules and techniques with local customs and agricultural cycles.



The execution of the recovery plan, where local governance systems, like the Panchayat, play a critical role.

IMPLEMENTATION

Engage the Community in Monitoring and Evaluation Processes: Create feedback mechanisms that allow the community to report on shelter effectiveness and suggest improvements.



MONITORING EVALUATION

This phase involves continuously assessing the effectiveness of the recovery and reconstruction activities. It ensures that the planned interventions are being implemented as intended and that they are achieving the desired impact.

IMPLEMENTATION IN RAJANPUR

Actively engage the basti walay (village community members) in rebuilding efforts to strengthen ownership and ensure the sustainability of the shelters. This approach will promote long-term maintenance and create a sense of belonging among residents.

Organize taamiri din (community building days) where neighbors help each other construct shelters, reinforcing community bonds and collective resilience. Encourage activities like haathi lagana (group lifting for construction tasks) to build solidarity.

IMPLEMENTATION IN RAJANPUR

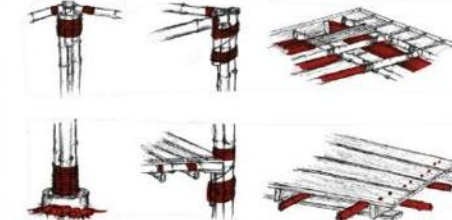
Develop raye shumari (feedback mechanisms) that enable community members to report on the effectiveness of the shelters and propose improvements. This could include regular meetings, suggestion boxes, and mobile surveys.

Form a muqami nigran council (local monitoring council) comprising representatives from different groups within the community, including women, elders, artisans, and youth, to oversee the evaluation process. This council should meet regularly to assess progress, identify challenges, and discuss necessary adjustments.

Utilize local labor for construction using familiar techniques such as kachcha (mud walls) and sarai (thatch roofs), which are not only cost-effective but also offer excellent thermal insulation, keeping homes cool in summer and warm in winter.

VERNACULAR METHODS & MATERIALS

bamboo framing and joinery details



Gara plaster (mud mixed with lime or choona)



kachcha eent (mud-brick) chhajje (thatch roofing)



deewan khana (courtyard design)

Maintain the traditional deewan khana (courtyard design), which serves as a communal space for social gatherings, food drying, and cultural rituals like mehndi or chowk pooja. To protect against flooding, these angans (courtyards) should be raised slightly on miti bharai (earth fill) foundations, which have been used effectively in local flood-prone areas.

Organize taamiri din (community building days) where neighbors help each other construct shelters, reinforcing community bonds and collective resilience. Encourage activities like haathi lagana (group lifting for construction tasks) to build solidarity.

TOOL

Consult the zila ke buzurg (community elders) and refer to the lokri zarait taqweem (local agricultural calendar) to time building activities in a way that aligns with traditional practices and maximizes community involvement.



TOOL

Use gaon ki jalsa (village gatherings) and local media, such as ghar ghar nigrani patrika (community newsletters) or dhandhora (announcements via loudspeakers), to share updates on the progress, challenges, and successes of the shelter projects. This will ensure transparency, maintain trust, and encourage continuous community engagement and feedback.

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“WAY FORWARD”

- Further investigation
- End Note
- Questions Answers

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WAY FORWARD

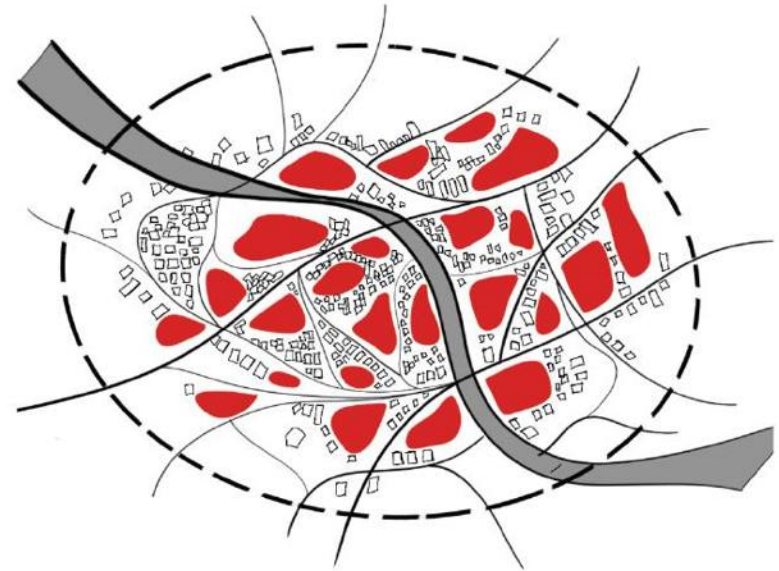
STREAMLINED PATH FOR COMMUNITY-INTEGRATED TRANSITIONAL HOUSING

TO EFFECTIVELY INTEGRATE COMMUNITY DYNAMICS into transitional housing solutions after disasters, it is essential to prioritize community engagement throughout the planning and construction phases. This approach ensures that housing designs are culturally, socially, and politically appropriate, leading to better adoption and adaptation by the affected communities.

INVESTIGATING THE CULTURAL RELEVANCE OF ARCHITECTURAL STYLES AND SPATIAL ORGANIZATION will be key in enhancing the acceptance of these housing solutions. The research should emphasize the economic viability of utilizing local labor and materials, which not only stimulates the local economy but also enhances the sustainability of the housing projects.

SCALABLE AND MODULAR DESIGNS should be explored to allow for flexibility in meeting the evolving needs of a recovering community, serving as an effective bridge to permanent housing. Assessing the subjective experiences of the inhabitants through qualitative methods will be crucial in evaluating the effectiveness of these housing solutions, particularly regarding user satisfaction, functionality, and the role of communal spaces in fostering community cohesion.

Finally, a **COLLABORATIVE FRAMEWORK SHOULD BE PROPOSED**, involving universities, NGOs, and governments, to ensure that ongoing academic research is translated into practical applications in the field. Simplifying this comprehensive research into practical guidelines will aid field workers in implementing effective transitional housing strategies. This approach will ensure that transitional housing is not only a temporary solution but a step towards a resilient and sustainable recovery process for communities like those in the Rajanpur District.



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